

CUSTOMER CHURN PREDICTION

Telecom Customer Churn Analysis & Machine Learning

Professional Project Report

Prepared for Recruiter / Interview Presentation

Project KPI	Result
Customer records	7,043
Original variables	50
Final model	Logistic Regression
ROC-AUC	0.9002

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GitHub: <https://github.com/jahidkhan911/customer-churn-prediction>

1. Executive Summary

This project develops an end-to-end machine learning workflow for telecom customer churn prediction using 7,043 customer records and 50 original variables.

Three classification algorithms were evaluated. After removing leakage-prone variables, Logistic Regression was selected as the final model with 83.53% accuracy, 67.38% churn recall, 0.6855 churn F1-score, and 0.9002 ROC-AUC.

A central quality-control lesson was data leakage prevention. Customer Status, Churn Score, Churn Category, Churn Reason, and Satisfaction Score were excluded because they could provide information unavailable at prediction time.

2. Business Problem

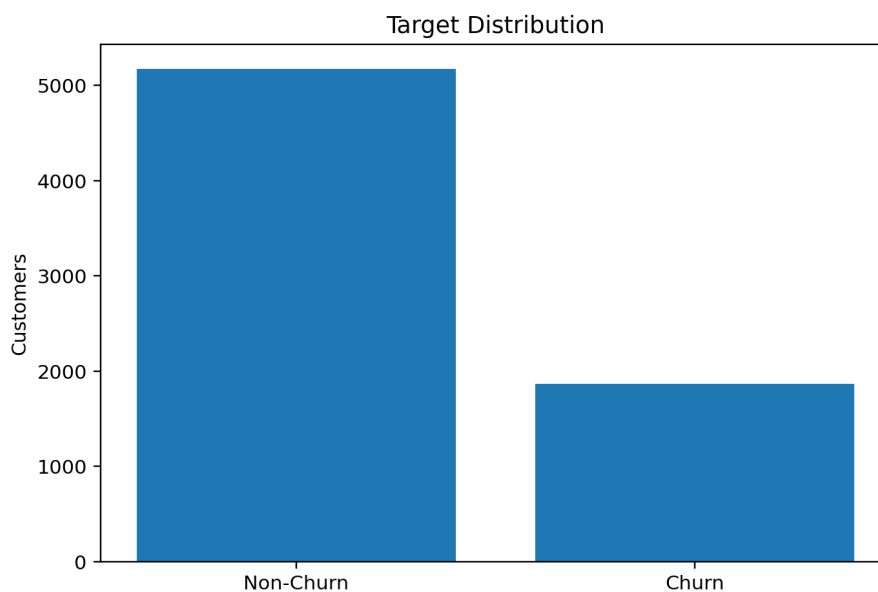
Telecom businesses need to identify customers at higher risk of leaving before churn occurs. Predictive churn scoring can support proactive retention by prioritizing customers for engagement, support, or appropriate retention offers.

Business question: Can historical customer information be used to identify customers at higher risk of churn and understand the factors associated with customer attrition?

3. Dataset & Target

The dataset contains 7,043 customers and 50 original columns. The target is Churn Label.

Target	Customers	Share
No Churn	5,174	73.5%
Churn	1,869	26.5%



4. Analytical Methodology

Data understanding and EDA; data cleaning; missing-value treatment; feature selection; leakage prevention; categorical encoding; stratified 80/20 train-test split; StandardScaler fitted only on training data; model training; evaluation; coefficient analysis; business interpretation.

5. Data Quality & Leakage Prevention

Customer ID, City, Zip Code, Latitude, Longitude, Quarter, and Population were removed from the modeling approach for identifier, cardinality, geographic, or limited-variation reasons.

Customer Status, Churn Score, Churn Category, Churn Reason, and Satisfaction Score were treated as leakage-prone. The project initially produced unusually high performance while leakage-prone information was present; removing it produced a more realistic evaluation.

6. Data Preprocessing

Missing Internet Type and Offer values were mapped to business-consistent categories such as 'No internet' and 'No offer'. Binary categorical variables were label encoded, while multi-category variables including Internet Type, Contract, and Payment Method were one-hot encoded with drop_first=True.

The dataset used an 80/20 stratified split. StandardScaler was fitted on training data only and then applied to test data.

7. Machine Learning Models

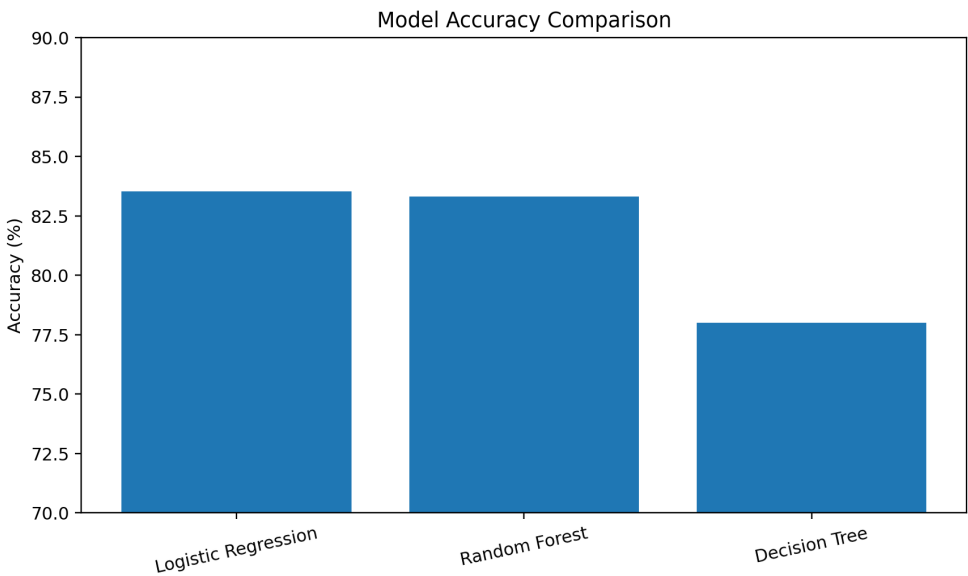
Logistic Regression — interpretable baseline and coefficient analysis.

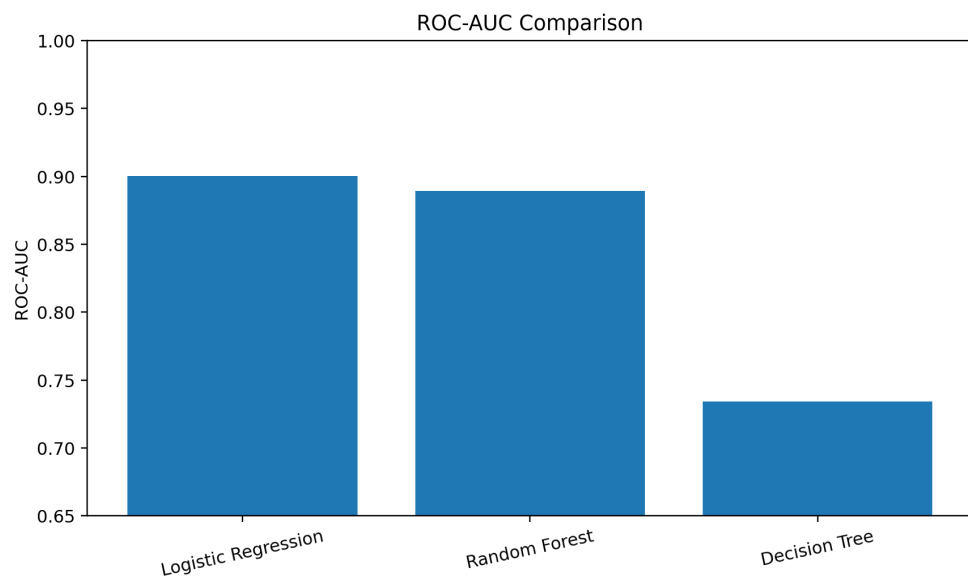
Decision Tree — nonlinear decision rules.

Random Forest — ensemble of decision trees.

8. Model Performance

Model	Accuracy	ROC-AUC	Churn Recall	Churn F1
Logistic Regression	83.53%	0.9002	67.38%	0.6855
Random Forest	83.32%	0.8894	61.00%	0.6600
Decision Tree	78.00%	0.7341	64.00%	0.6100





9. Model Interpretation & Key Findings

The strongest positive coefficient was Monthly Charge (+1.848), while Number of Referrals (-1.782), Two-Year Contract (-1.181), and Tenure in Months (-0.932) showed strong negative associations with predicted churn.

These are statistical associations, not proof of causation.

10. Business Insights & Recommended Actions

Higher monthly charges: review perceived value and pricing for high-charge customers.

Shorter tenure: strengthen onboarding and early engagement.

Longer-term contracts: consider suitable incentives and added value.

Referral activity: strengthen referral and loyalty programs.

Dependents: evaluate family-oriented plans and household benefits.

Security and technical support: evaluate appropriate value/retention packages.

High-risk customers: prioritize proactive support, personalized offers, and retention campaigns.

11. Why This Project Demonstrates Business Analytics Capability

Combines business problem definition with an end-to-end analytical workflow.

Demonstrates Python, Pandas, and Scikit-learn.

Shows data-quality reasoning and explicit leakage prevention.

Uses multiple metrics appropriate for an imbalanced classification problem.

Connects model outputs to customer-retention actions.

12. Limitations

Historical data may not perfectly represent future behavior; relationships are statistical associations; leakage-prone variables were removed; the current model uses a standard threshold; further optimization could reflect business costs of prediction errors.

13. Future Improvements

Hyperparameter optimization, stratified cross-validation, threshold optimization, SHAP/LIME explainability, a production preprocessing pipeline, and Streamlit/Flask/FastAPI deployment.

14. 60-Second Recruiter / Interview Summary

I built an end-to-end telecom customer churn prediction project using 7,043 customer records. I performed EDA and data-quality checks, handled categorical variables, removed irrelevant and leakage-prone features, and compared Logistic Regression, Decision Tree, and Random Forest. Logistic Regression was selected with 83.53% accuracy and 0.9002 ROC-AUC. A key part of the project was identifying leakage from variables such as Churn Reason, Churn Category, Customer Status, Churn Score, and Satisfaction Score. I then interpreted model coefficients and translated the findings into retention actions around monthly charges, tenure, contracts, referrals, and customer support.

15. Source & Repository

Project repository: <https://github.com/jahidkhan911/customer-churn-prediction>

This report is based on the repository README, notebook workflow, reported model metrics, and feature-analysis results available in the repository as of September 2026.